

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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2026-08-31

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы

```
miradov@miradov:~$ touch abc1
miradov@miradov:~$ cp abc1 april
miradov@miradov:~$ cp abc1 may
miradov@miradov:~$ mkdir monthly
miradov@miradov:~$ cp april may monthly
miradov@miradov:~$ cp monthly/may monthly/june
miradov@miradov:~$ ls monthly
april  june  may
miradov@miradov:~$ mkdir monthly.00
miradov@miradov:~$ cp -r monthly monthly.00
miradov@miradov:~$ cp -r monthly.00 /tmp
miradov@miradov:~$
```

Рисунок 1: Выполнение примеров

```
miradov@miradov:~$ mv april july
miradov@miradov:~$ mv july monthly.00
miradov@miradov:~$ ls monthly.00
july  monthly
miradov@miradov:~$ mv monthly.00 monthly.01
miradov@miradov:~$ mkdir reports
miradov@miradov:~$ mv monthly.01 reports
miradov@miradov:~$ mv reports/monthly.01 reports/monthly
miradov@miradov:~$
```

Рисунок 2: Выполнение примеров

```
miradov@miradov:~$  
miradov@miradov:~$ touch may  
miradov@miradov:~$ ls -l may  
-rw-r--r--. 1 miradov miradov 0 Aug 31 11:04 may  
miradov@miradov:~$ chmod u+x may  
miradov@miradov:~$ ls -l may  
-rwxr--r--. 1 miradov miradov 0 Aug 31 11:04 may  
miradov@miradov:~$ chmod u-x may  
miradov@miradov:~$ ls -l may  
-rw-r--r--. 1 miradov miradov 0 Aug 31 11:04 may  
miradov@miradov:~$ mkdir monthly  
mkdir: cannot create directory 'monthly': File exists  
miradov@miradov:~$ chmod g-r,o-r monthly  
miradov@miradov:~$ touch abc1  
miradov@miradov:~$ chmod g+w abc1  
miradov@miradov:~$
```

Рисунок 3: Выполнение примеров

Создание директорий и копирование файлов

```
miradov@miradov:~$  
miradov@miradov:~$ cp /usr/include/linux/sysinfo.h ~  
miradov@miradov:~$ mv sysinfo.h equipment  
miradov@miradov:~$ mkdir ski.plases  
miradov@miradov:~$ mv equipment ski.plases/  
miradov@miradov:~$ mv ski.plases/equipment ski.plases/equiplist  
miradov@miradov:~$ touch abc1  
miradov@miradov:~$ cp abc1 ski.plases/equiplist2  
miradov@miradov:~$ cd ski.plases/  
miradov@miradov:~/ski.plases$ mkdir equipment  
miradov@miradov:~/ski.plases$ mv equiplist equipment/  
miradov@miradov:~/ski.plases$ mv equiplist2 equipment/  
miradov@miradov:~/ski.plases$ cd  
miradov@miradov:~$ mkdir newdir  
miradov@miradov:~$ mv newdir ski.plases/  
miradov@miradov:~$ mv ski.plases/newdir/ ski.plases/plans  
miradov@miradov:~$
```

Рисунок 4: Работа с каталогами

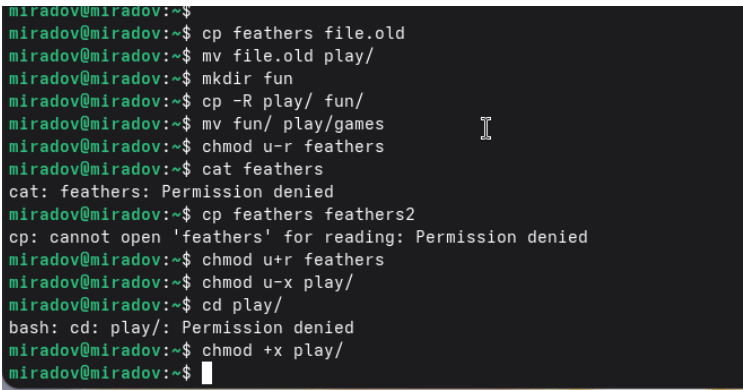
Работа с командой chmod

```
miradov@miradov:~$ mkdir australia play
miradov@miradov:~$ touch my_os feathers
miradov@miradov:~$ chmod 744 australia/
miradov@miradov:~$ chmod 711 play/
miradov@miradov:~$ chmod 544 my_os
miradov@miradov:~$ chmod 664 feathers
miradov@miradov:~$ ls -l
total 0
-rw-rw-r--. 1 miradov miradov  0 Aug 31 11:06 abc1
drwxr--r--. 1 miradov miradov  0 Aug 31 11:07 australia
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Desktop
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Documents
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Downloads
-rw-rw-r--. 1 miradov miradov  0 Aug 31 11:07 feathers
drwxr-xr-x. 1 miradov miradov 74 Aug 31 10:31 git-extended
-rw-r--r--. 1 miradov miradov  0 Aug 31 11:04 may
drwx--x--x. 1 miradov miradov 24 Aug 31 11:03 monthly
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Music
-r-xr--r--. 1 miradov miradov  0 Aug 31 11:07 my_os
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Pictures
drwx--x--x. 1 miradov miradov  0 Aug 31 11:07 play
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Public
drwxr-xr-x. 1 miradov miradov 14 Aug 31 11:03 reports
drwxr-xr-x. 1 miradov miradov 28 Aug 31 11:06 ski.places
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Templates
drwxr-xr-x. 1 miradov miradov  0 Aug 31 09:53 Videos
drwxr-xr-x. 1 miradov miradov 10 Aug 31 10:03 work
```

```

root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
openvpn:x:990:990:OpenVPN:/etc/openvpn:/usr/sbin/nologin
nm-openvpn:x:989:989:Default user for running openvpn spawned by NetworkManager:/:/usr/sbin/nologin
abrt:x:173:173:/:etc/abrt:/usr/sbin/nologin

```



```
miradov@miradov:~$  
miradov@miradov:~$ cp feathers file.old  
miradov@miradov:~$ mv file.old play/  
miradov@miradov:~$ mkdir fun  
miradov@miradov:~$ cp -R play/ fun/  
miradov@miradov:~$ mv fun/ play/games  
miradov@miradov:~$ chmod u-r feathers  
miradov@miradov:~$ cat feathers  
cat: feathers: Permission denied  
miradov@miradov:~$ cp feathers feathers2  
cp: cannot open 'feathers' for reading: Permission denied  
miradov@miradov:~$ chmod u+r feathers  
miradov@miradov:~$ chmod u-x play/  
miradov@miradov:~$ cd play/  
bash: cd: play/: Permission denied  
miradov@miradov:~$ chmod +x play/  
miradov@miradov:~$
```

Рисунок 7: Работа с файлами и правами доступа

```
MOUNT(8)                                System Administration                                MOUNT(8)

NAME
    mount - mount a filesystem

SYNOPSIS
    mount [-h|-V]

    mount [-l] [-t fstype]

    mount -a [-fFnrsvw] [-t fstype] [-O optlist]

    mount [-fnrsvw] [-o options] device | mountpoint

    mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

    mount --bind|--rbind|--move olddir newdir

    mount
    --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable]
]
    mountpoint

DESCRIPTION
    All files accessible in a Unix system are arranged in one big tree, the
    file hierarchy, rooted at /. These files can be spread out over several
    devices. The mount command serves to attach the filesystem found on some
    device to the big file tree. Conversely, the umount(8) command will detach
    it again. The filesystem is used to control how data is stored on the
    device or provided in a virtual way by network or other services.

    The standard form of the mount command is:

    mount -t type device dir
```

```
FCK(8)                                System Administration                                FCK(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--]
    [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems.
    filesystem can be a device name (e.g., /dev/hdc1, /dev/sdb2), a mount
    point (e.g., /, /usr, /home), or a filesystem label or UUID specifier
    (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root). Normally,
    the fsck program will try to handle filesystems on different physical disk
    drives in parallel to reduce the total amount of time needed to check all
    of them.

    If no filesystems are specified on the command line, and the -A option is
    not specified, fsck will default to checking filesystems in /etc/fstab
    serially. This is equivalent to the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0
      No errors

    1
      Filesystem errors corrected

    2
      System should be rebooted

    4
```

```

MKFS(8)                                     System Administration      MKFS(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific
    mkfs.<type> utils.

    mkfs is used to build a linux filesystem on a device, usually a hard disk
    partition. The device argument is either the device name (e.g., /dev/hda1,
    /dev/sdb2), or a regular file that shall contain the filesystem. The size
    argument is the number of blocks to be used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem
    builders (mkfs.fstype) available under Linux. The filesystem-specific
    builder is searched for via your PATH environment setting only. Please see
    the filesystem-specific builder manual pages for further details.

OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the
        default filesystem type (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem
        builder.

    -V, --verbose
```

```
KILL(1)                                User Commands                                KILL(1)

NAME
    kill - terminate a process

SYNOPSIS
    kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds
signal] [--] pid|name...

    kill -l [number|0xsigmask] | -L

    kill -d pid

DESCRIPTION
    The command kill sends the specified signal to the specified processes or
    process groups.

    If no signal is specified, the TERM signal is sent. The default action for
    this signal is to terminate the process. This signal should be used in
    preference to the KILL signal (number 9), since a process may install a
    handler for the TERM signal in order to perform clean-up steps before
    terminating in an orderly fashion. If a process does not terminate after a
    TERM signal has been sent, then the KILL signal may be used; be aware that
    the latter signal cannot be caught, and so does not give the target
    process the opportunity to perform any clean-up before terminating.

    Most modern shells have a builtin kill command, with a usage rather
    similar to that of the command described here. The --all, --pid, and
    --queue options, and the possibility to specify processes by command name,
    are local extensions.

    If signal is 0, then no actual signal is sent, but error checking is still
    performed.
```

3. Выводы по проделанной работе

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.